MSG No. 102.194 P_I4_2nm [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'm'm`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,4,7,8,10,11,13,14]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[2,3,5,6,9,12,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2m'm'.`

No.	position	mapping
1	$\left[0,\, \frac{1}{2},\, z\right]$	`[1,4,13,14]`
2	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[2,3,15,16]`
3	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	`[5,6,9,12]`
4	$\left[\frac{1}{2},\, 0,\, z\right]$	`[7,8,10,11]`

Table 3: Wyckoff site: `8c`, site symmetry: `..m`

No.	position	mapping
1	$[x,\, x,\, z]$	`[1,8]`
2	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[2,5]`
3	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[3,6]`
4	$[-x,\, -x,\, z]$	`[4,7]`
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9,16]`
6	$[-x,\, x,\, z]$	`[10,13]`
7	$[x,\, -x,\, z]$	`[11,14]`
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[12,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\, 0,\, z]$	`[1,14]`
2	$\left[\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[2,16]`
3	$\left[\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[3,15]`
4	$[-x,\, 0,\, z]$	`[4,13]`
5	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[5,12]`
6	$\left[x+\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[6,9]`
7	$[0,\, -x,\, z]$	`[7,11]`
8	$[0,\, x,\, z]$	`[8,10]`

Table 5: Wyckoff site: `16e`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[2]`

continued ...

Table 5

No.	position	mapping
3	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[5]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[6]
7	$[-y,\ -x,\ z]$	[7]
8	$[y,\ x,\ z]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$[-y,\ x,\ z]$	[10]
11	$[y,\ -x,\ z]$	[11]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$[-x,\ y,\ z]$	[13]
14	$[x,\ -y,\ z]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]